JAVA CONSOLE APPLICATION

Weekly Meal Planner & Grocery Generator

A practical console tool for planning meals and tracking grocery spending.



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COURSE OUTCOMES: CO1, CO2, AND CO3

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01 · OVERVIEW

Plan the week before you shop

One weekly plan reduces last-minute decisions and keeps family grocery spending visible.

Organize meals

Build a 7-day meal plan before shopping.

Generate groceries

Turn meals into items, quantities, prices, and totals.

Check the budget

Calculate costs and keep spending visible.

Find an item

Search the grocery list without extra libraries.



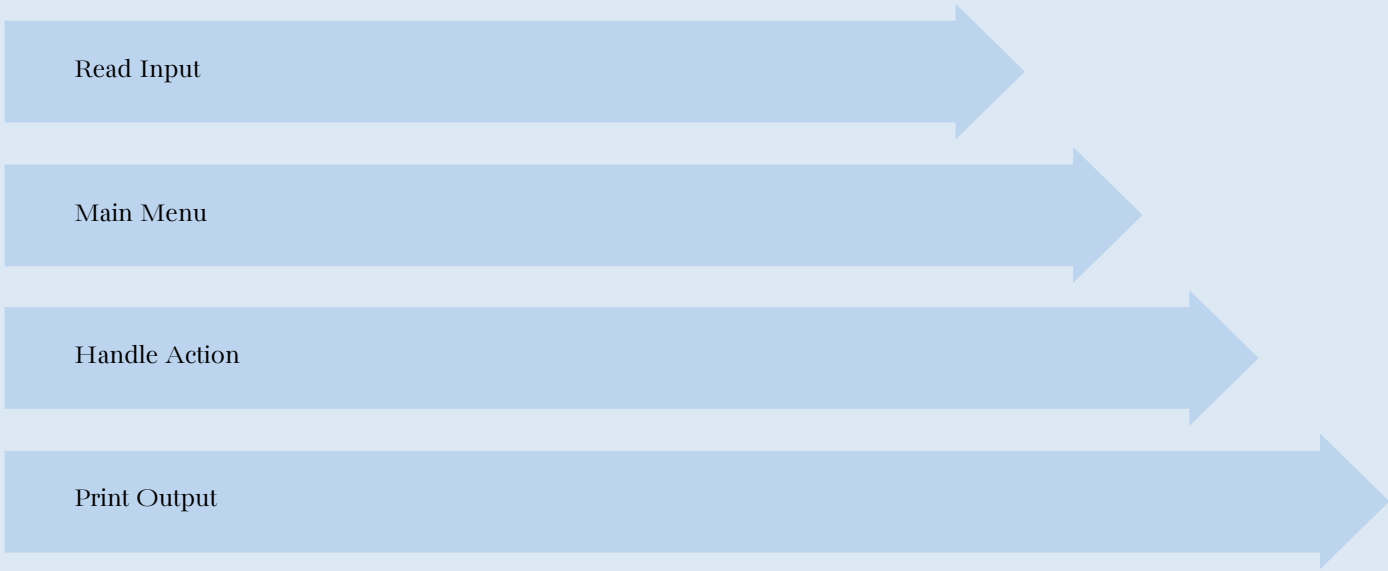
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02 · JAVA STRUCTURE

The program starts with a simple Java structure

Input enters through the console, methods process it, and results return as formatted output.

VARIABLES · SCANNER · MAIN() · DO-WHILE



Types

String, int, double, boolean

Flow

input → menu choice → method → result

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03 · PROFILE

A profile turns preferences into decisions

Capture: Enter `userName` and `familyMembers` with `Scanner`.

Budget: Set `weeklyBudget` as a decimal value for accurate cost calculations.

Preference: Select a food preference: 1 = Vegetarian, 2 = Non-Vegetarian, 3 = Mixed.

Translate: Use if-else logic to translate the choice into a readable profile label.

RELEVANT METHODS: ENTERUSERDETAILS(), SELECTFOODPREFERENCE(), AND DISPLAYUSERPROFILE().

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04 · MEAL PLAN

The meal plan maps 7 days to 3 meals

Rows represent days. Columns represent meal slots. Nested loops populate all 21 cells.

Day	Breakfast	Lunch	Dinner	
Monday	—	—	—	
Tuesday	—	—	—	
Wednesday	—	—	—	
Thursday	—	—	—	
Friday	—	—	—	
Saturday	—	—	—	
Sunday	—	—	—	

```
STRING[][] MEALPLAN = NEW STRING[7][3];
```

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05 · GROCERY VIEW

The grocery list turns meals into a shopping view

Parallel arrays keep each item aligned from name to total cost.

Item	Quantity	Price
groceryItems[]	quantities[]	prices[]

Calculate and display

A for loop calculates each item cost and totalGroceryCost, then formats the console table.

ITEM · QUANTITY · PRICE · TOTAL

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06 · BUDGET

Budget management makes the total meaningful

A grocery total becomes a decision when it is compared with the weekly budget.

LOW

Less than 20% remaining

MEDIUM

Less than 50% remaining

HIGH

At least 50% remaining

REMAINING BUDGET = WEEKLYBUDGET − TOTAL EXPENSES

Arithmetic makes the status visible

`+` `-` `*` `/`

Comparisons such as `<`, `>`, and `==` classify the result with if-else-if conditions.

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07 · SEARCH

Linear search finds an item without extra libraries

The method checks each grocery item in order and stops at the first match.

01

Ask

Request a grocery item name.

02

Loop

Visit groceryItems[] from first to last.

03

Compare

Use equalsIgnoreCase().

04

Report

Set found = true and break, or report “not found”.

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08 · COURSE OUTCOMES

CO1, CO2, and CO3 work together

The same program flow turns fundamentals into a working result.

CAPTURE INPUT → CHOOSE A CASE → PROCESS → DISPLAY A RESULT

CO1

Variables, data types, arithmetic, comparison, and assignment operators.

CO2

if-else, if-else-if, switch-case, for, nested for, and do-while statements.

CO3

Reusable methods, 1D arrays for grocery data, and the 2D mealPlan array.

```
Scanner scanner = new Scanner(System.in);

String[][] mealPlan = new String[7][3];

searchGroceryItem()
```

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09 · CLOSE

A practical console tool with room to grow

Benefits

Saves time in weekly planning, organizes shopping, and keeps family expenses visible.

Achievement

Connects Java fundamentals to a working meal, grocery, search, and budget workflow.

Future scope

Add nutritional information, recipe recommendations, and a graphical user interface.



THANK YOU — QUESTIONS ARE WELCOME.

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